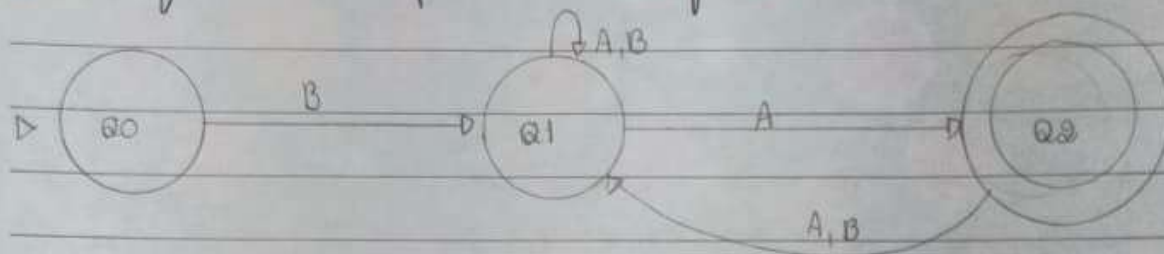


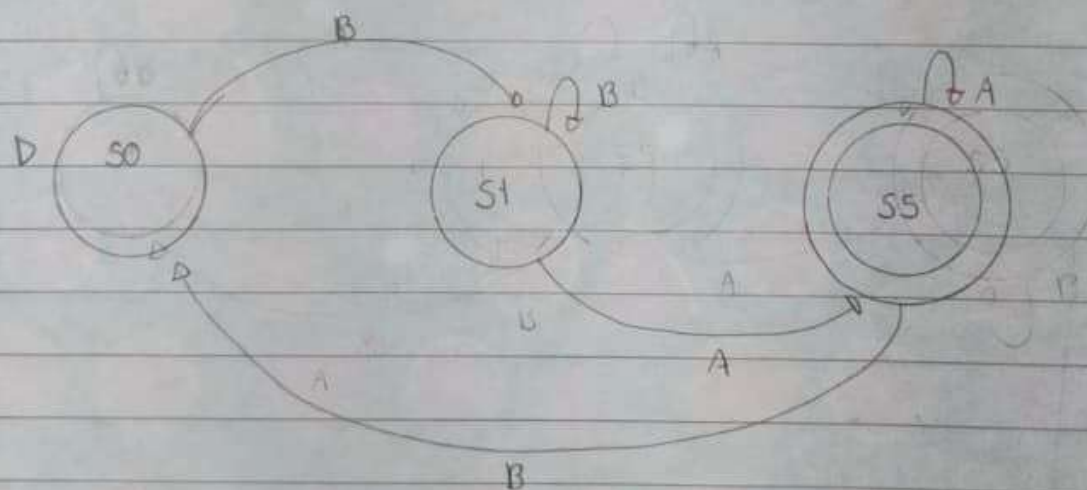
Unidades AFD e AFN - Introdução à Teoria da Computação

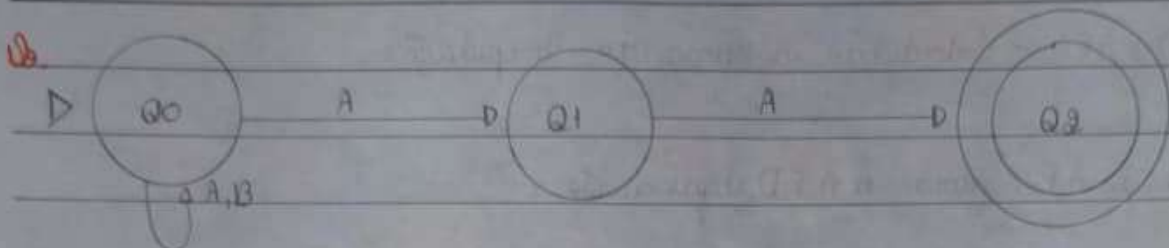
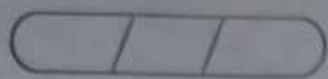
1. Transforme o AFN para o AFD equivalente.



S	A	B	S	va	vb
Q0	-	Q1	- S0	Q0	Q1 S1
Q1	Q1, Q2	Q1	- S1	Q1	Q1, Q2 S5
Q2	Q1	Q1	- S2	Q2	Q1 S1
			- S3	Q0, Q1	Q1, Q2 S5
			- S4	Q0, Q2	Q1 S1
			* S5	Q2, Q1	Q1, Q1 S1
			- S6	Q0, Q1, Q2	Q1, Q2, Q1 S1
			- S7	∅	-

$$2^3 = 8$$

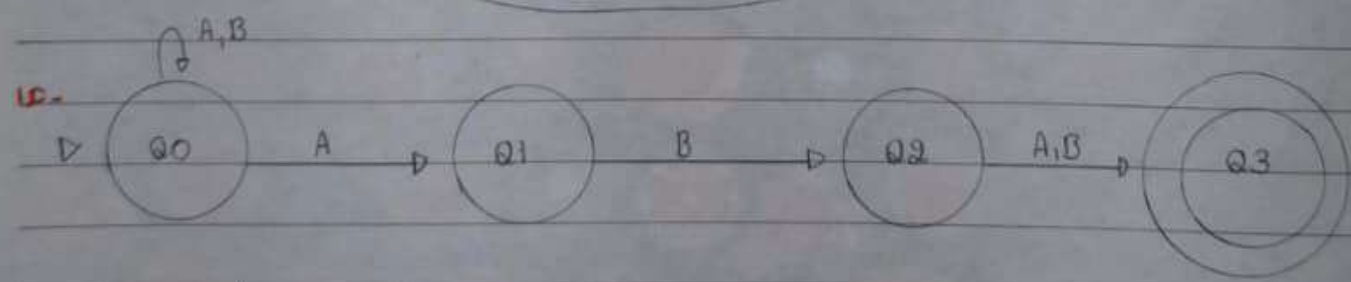
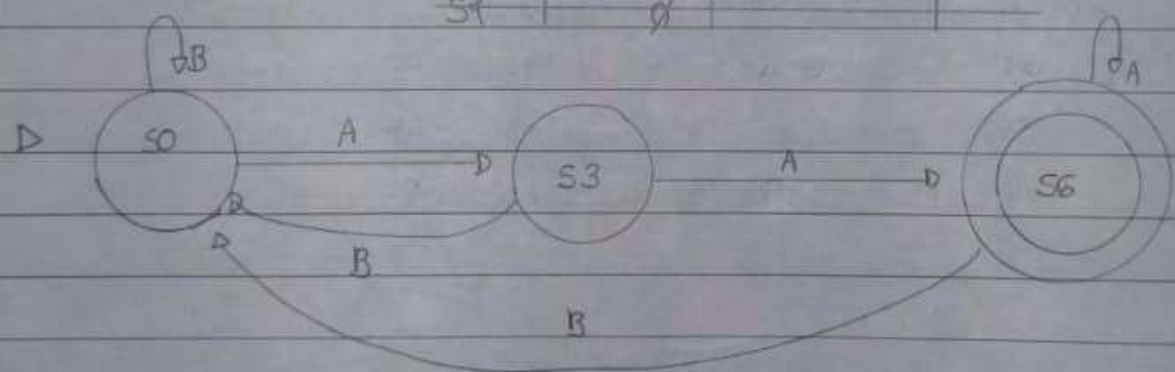




δ	A	B	δ	A	B
Q0	{Q0, Q1}	{Q0}	S0	{Q0, Q1, S3}	{Q0, S0}
Q1	{Q2}	-	S1	{Q1, Q2, S2}	-
Q3	-	-	S2	{Q2}	-

$2^3 = 8$

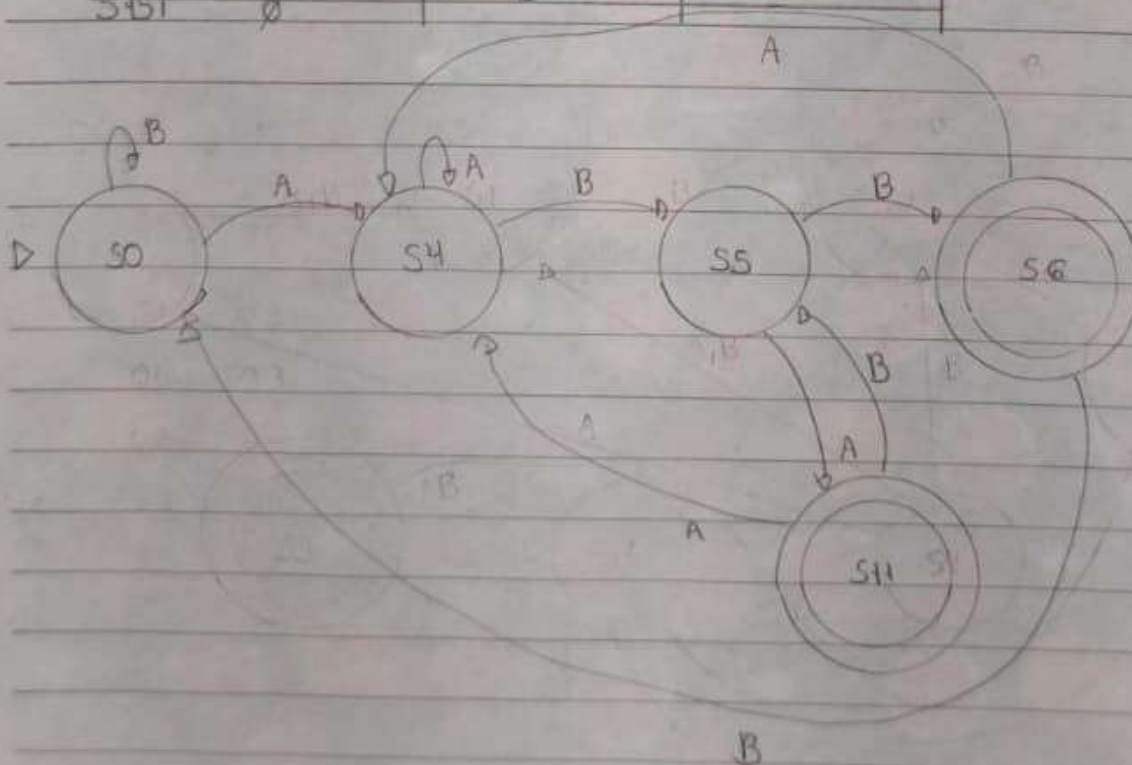
S3	{Q0, Q1, Q2, S6}	{Q0, S0}
S4	{Q0, Q1, Q2, S3}	-
S5	{Q1, Q2, S5}	-
S6	{Q0, Q1, Q2, S6}	{Q0, S0}
S7	$\emptyset$	-

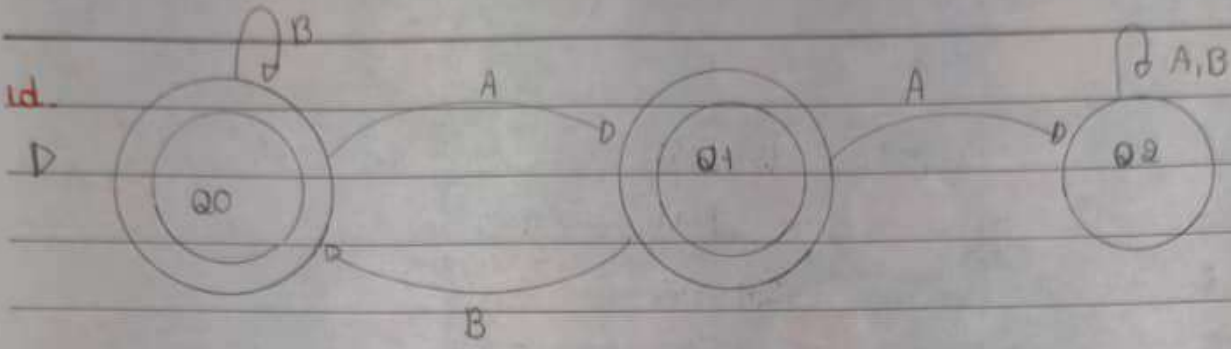
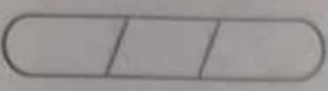


δ	A	B
Q0	{Q0, Q1}	{Q0}
Q1	-	{Q2}
Q2	{Q3}	{Q3}
Q3	-	-

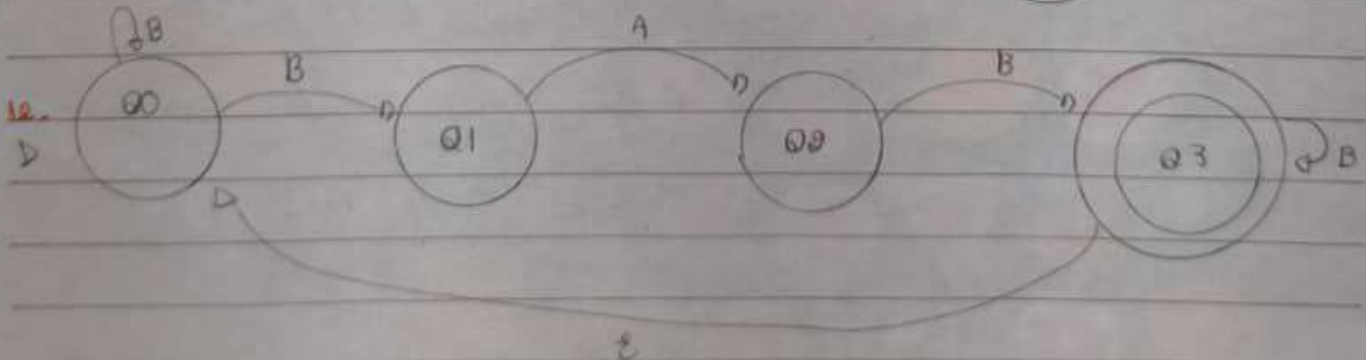
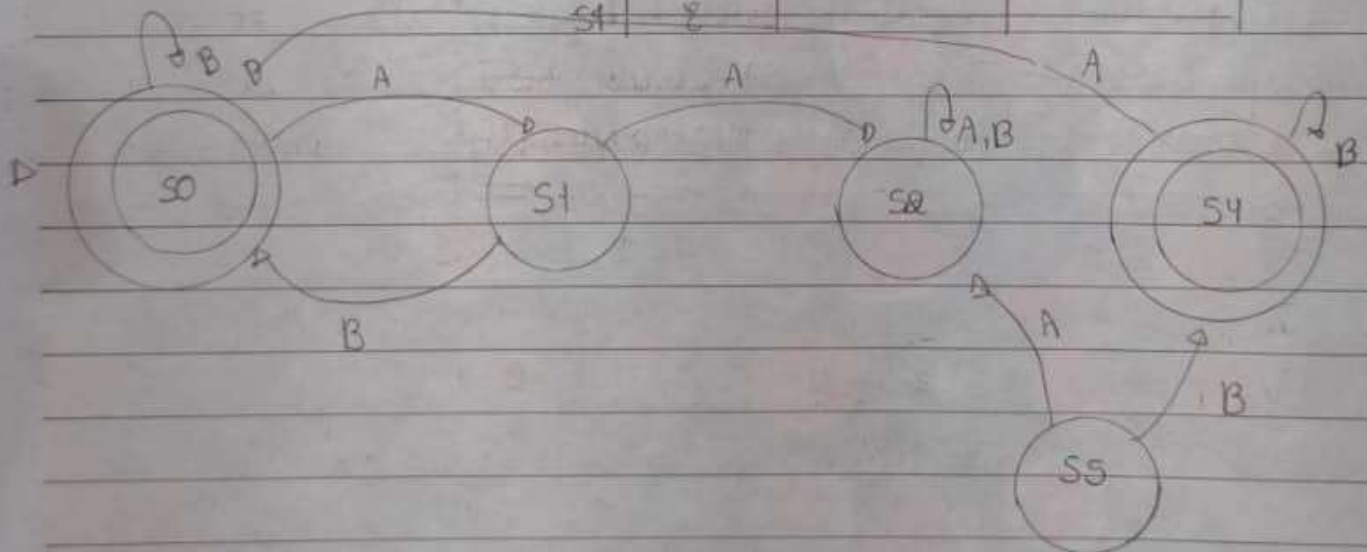
$2^4 = 16$

	S	A	B
S0	00	00, 01 (54)	00 (50)
S1	01	-	02 (52)
S2	02	03 (53)	03 (53)
* S3	03	-	-
S4	00, 01	00, 01 (54)	00, 02 (55)
S5	00, 02	00, 01, 03 (511)	00, 03 (56)
* S6	00, 03	00, 01 (54)	00 (50)
S7	01, 02	03 (53)	02, 03 (53)
S8	01, 03	-	03 (53)
S9	02, 03	03 (53)	03 (53)
S10	00, 01, 02	00, 01, 03 (511)	00, 02, 03 (512)
* S11	00, 01, 03	00, 01 (54)	00, 02 (55)
S12	00, 02, 03	00, 01, 03 (511)	00, 03 (56)
S13	01, 02, 03	03 (53)	02, 03 (53)
S14	00, 01, 02, 03	00, 01, 03 (511)	00, 02, 03 (512)
S15	∅	-	-

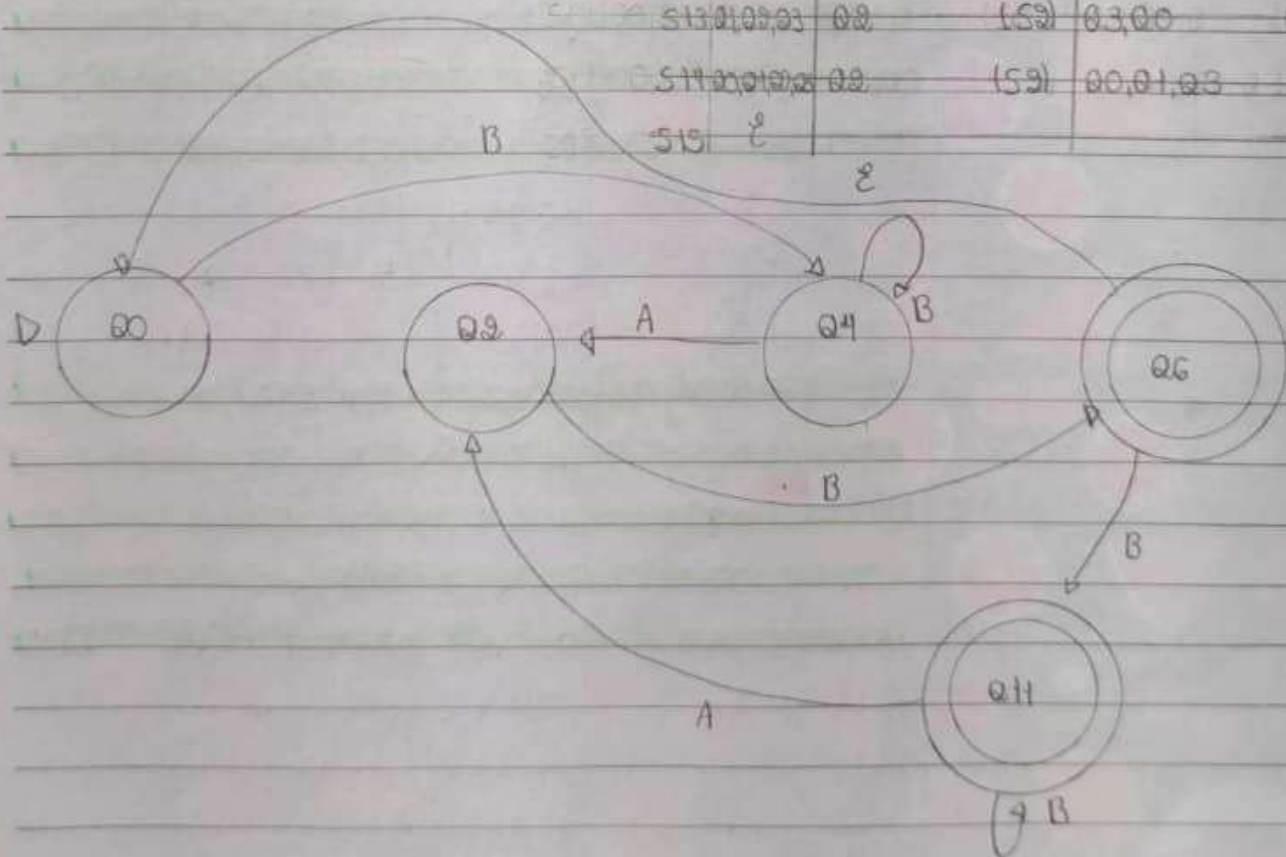


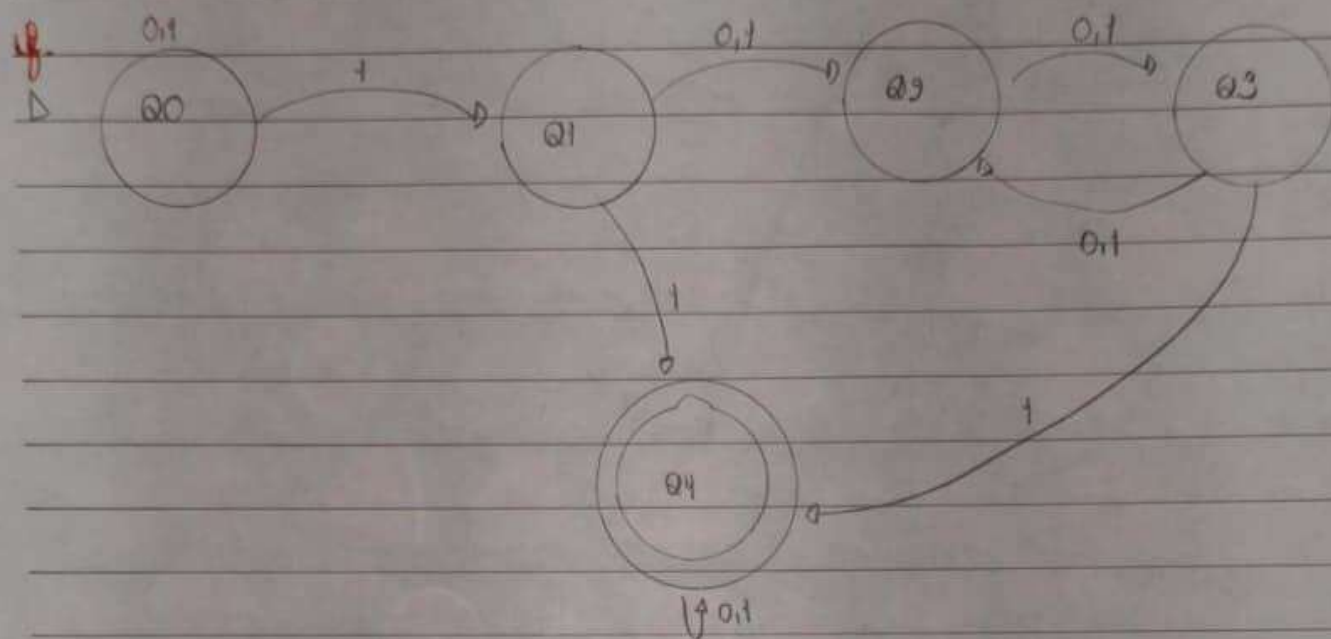
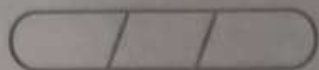


S	A	B		S	A	B
Q0	Q1	Q0	* S0	Q0	Q1 S1	Q0 S0
Q1	Q2	Q0	S1	Q1	Q2 S2	Q0 S0
Q2	Q2	Q2	S2	Q2	Q2 S2	Q2 S2
			S3	Q0, Q1	Q1, Q2 S5	Q0, Q0 S0
			* S4	Q0, Q2	Q1, Q2 S5	Q0, Q2 S4
			S5	Q1, Q2	Q2, Q2 S2	Q0, Q2 S4
			S6	Q0, Q1, Q2	Q0, Q2, Q2 S4	Q0, Q0, Q2 S4
			S1	ε		

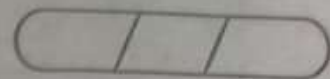


S	A	B	ϵ	S	A	B
Q0	-	{Q0, Q1}	-	S0 {Q0}	-	Q0, Q1 (S4)
Q1	{Q2}	-	-	S1 {Q1}	Q2 (S2)	-
Q2	-	{Q3}	-	S2 {Q2}	-	Q3, Q0 (S6)
Q3	-	{Q3}	{Q0}	S3 {Q3}	-	Q3, Q0 (S6)
				S4 {Q0, Q1}	Q2 (S2)	Q0, Q1 (S4)
Joda vez que temos um Q3 voltamos Q0				S5 {Q0, Q2}	-	Q0, Q1, Q3 (S11)
				* S6 {Q0, Q3}	-	Q0, Q1, Q3 (S11)
				S7 {Q1, Q2}	Q2	Q3, Q0
				S8 {Q1, Q3}	Q2 (S2)	Q3, Q0 (S6)
				S9 {Q2, Q3}	-	Q3, Q0 (S6)
				S10 {Q0, Q1, Q2}	Q2 (S2)	Q0, Q1, Q3 (S11)
				* S11 {Q0, Q1, Q3}	Q2 (S2)	Q0, Q1, Q3 (S11)
				S12 {Q0, Q2}	-	Q0, Q1, Q3 (S11)
				S13 {Q1, Q2, Q3}	Q2 (S2)	Q3, Q0 (S6)
				S14 {Q0, Q1, Q2}	Q2 (S2)	Q0, Q1, Q3 (S11)
				S15 ϵ		





δ	0	1		δ	0	1
Q0	Q0	Q0, Q1	→	S1	Q0	Q0, Q1 (S1)
Q1	Q1, Q4	Q2		S2	Q1	Q2, Q4 (S2)
Q2	Q3	Q3		S3	Q2	Q3 (S3)
Q3	Q3	Q3, Q4		S4	Q3	Q3 (S4)
Q4	Q4	Q4		S5	Q4	Q4 (S5)
$2^5 = 32$			→	S6	Q0, Q1	Q0, Q2, Q4 (S6)
			→	S7	Q0, Q2	Q0, Q3 (S7)
			→	S8	Q0, Q3	Q0, Q3 (S8)
				S9	Q0, Q4	Q0, Q4 (S9)
				S10	Q1, Q2	Q2, Q4, Q3 (S10)
				S11	Q1, Q3	Q2, Q4, Q3 (S11)
				S12	Q1, Q4	Q2, Q4 (S12)
				S13	Q2, Q3	Q3, Q3, Q4 (S13)
			* →	S14	Q2, Q4	Q3, Q4 (S14)
			* →	S15	Q3, Q4	Q3, Q4, Q4 (S15)
			→	S16	Q0, Q1, Q2	Q0, Q2, Q4, Q3 (S16)
			→	S17	Q0, Q1, Q3	Q0, Q2, Q4, Q3 (S17)
				S18	Q0, Q1, Q4	Q0, Q2, Q4 (S18)



	S19 00,02,03	00,03,02 (S19)	00,01,03,02,04 (S31)
* →	S20 00,02,04	00,03,04 (S21)	00,01,03,04 (S28)
* →	S21 00,03,04	00,02,04 (S20)	00,01,02,04 (S29)
	S22 01,02,03	02,04,03 (S25)	02,03,04 (S25)
	S23 01,02,04	02,04,03 (S25)	02,03,04 (S25)
	S24 01,03,04	02,04 (S14)	02,01,04 (S14)
	S25 02,03,04	03,02,04 (S25)	03,02,04 (S25)
→	S26 00,01,02,03	00,02,04,03 (S29)	00,01,02,03,04 (S31)
* →	S27 00,01,02,04	00,02,04,03 (S29)	00,01,02,03,04 (S31)
* →	S28 00,01,03,04	00,02,04 (S20)	00,01,02,03,04 (S31)
* →	S29 00,02,03,04	00,03,02,04 (S20)	00,01,03,02,04 (S31)
	S30 01,02,03,04	02,04,03 (S25)	02,03,04 (S25)
* →	S31 00,01,02,03,04	00,02,04,03 (S29)	00,01,02,03,04 (S31)
	S32	ε	

